Scale or Shape? Decomposing NTK Movement During Grokking

Group G2
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Notes for the group. Delete this block before submission.

- **[TODO: Later, replace every mention of "rotation" with "shape", including R_t's name, the figure labels and the tier titles. The word "rotation" is kept for now so that the text matches what the group already uses.]**
- Each experiment section opens with its main finding. Every number comes from the scripts in `report_v2/code/` through `numbers.tex`. No red planning values are left, since the E14 cross-check was dropped on 17 October 2026.
- The main text must end by page 12. References and appendices do not count, and the appendices may take at most 5 pages. Each section shows its page budget.
- The draft notes after the consent page hold the criteria self-assessment, the changelog of what was cut and why, and the reference checks.
- Blue text marks the edits of 20 October 2026 that add the MNIST tier (section 4.5) and change the sections around it. A blue bracketed note marks each cut or move. Delete the blue markup after review, and keep the text.

Abstract

Grokking is delayed generalisation of a network, but what determines the delay is disputed. We decompose the movement of the neural tangent kernel into its scale and rotation, measuring both on a two-layer network learning modular addition across laziness, width and weight decay. At a fixed width, rotation is the statistic most tied to the grokking step, even though some networks reach it without grokking. The grokking time grows with width and falls with weight decay, which neither a rotation clock nor a weight-decay clock predicts alone. Read early, rotation is the best single predictor of the grokking time. We also measure the same statistics on MNIST, where grokking is induced in a different way by scaling up the initial weights. On this task the weight-decay clock holds and width drops out. Weight decay therefore sets the time through the weight norm on both tasks, and width enters only where the task forces the kernel to change shape.

1 Introduction

Page budget: 1.0 page.

Grokking is the delayed generalisation that Power et al. (2022) found on small algorithmic datasets. A network fits its training data early, and its test accuracy rises significantly long afterwards. Modular addition learned by a two-layer perceptron is a small instance of it. Full-batch gradient descent on the squared error groks there with no explicit regularisation (Gromov, 2023). What sets the length of the delay is an open question.

One account holds that grokking is a transition from lazy training to rich training (Kumar et al., 2024). In lazy training the network behaves as a kernel method with its initial tangent kernel. In rich training the kernel adapts to the task. The evidence that the kernel changes is a distance: the empirical tangent kernel stays close to its initial value until the training data are fitted, and moves far from it afterwards (Mohamadi et al., 2024). A distance does not say how the kernel changed. A kernel can grow in scale while its eigenvectors stay fixed. It can also rotate, so that its eigenvectors change.

The difference matters for the accounts of the grokking time. Recent work times grokking by scale-like parameters: the weight norm (Khanh et al., 2026), the scale of the logits (Khanh, 2026), or the weight-decay rate (Kim, 2026; Pracher et al., 2026). The result that Kumar et al. (2024) cite for weight decay forcing the kernel to change is that of Lewkowycz and Gur-Ari (2021), in which the kernel only shrinks. None of these accounts measures whether the kernel rotates.

Rotation is important for the lazy-to-rich account of grokking. A kernel that only grows keeps its eigenvectors, so a large distance does not by itself show that features were learned.

We ask what sets the grokking time: the shape of the kernel, its scale, or the weight-decay rate. Our hypothesis is that rotation (change in shape) sets it. We measure three statistics of the empirical tangent kernel along training: its scale S_t, its rotation R_t and its alignment A_t with the labels. We do so on a grid of 720 runs that crosses laziness, width and weight decay. Our contributions are as follows.

1. We split the movement of the kernel into a change of scale and a rotation, and use each one as a timing variable. On the baseline of Kumar et al. (2024), the usual distance is mostly a change of scale: at $\alpha = 1$ the scale accounts for 90% of D_t^2 at grokking.
2. We test whether grokking happens at a fixed value of each statistic, with a placebo test that reads each run at another run's grokking step, so that a statistic that merely stays in a narrow band is not mistaken for a threshold.
3. We fit a power law for how the grokking time scales with model width and weight decay. The hypotheses that rotation sets the grokking time and that weight decay sets it predict different exponents.
4. We test whether the statistics can be used to predict grokking ahead of time.
5. We repeat the first two tests on the MNIST dataset, a task on which Liu et al. (2023) induce grokking through a different method, to test if the findings are generalisable.

On modular addition, rotation is the most important single statistic, but it is not enough on its own: the grokking time depends on both width and weight decay. While on MNIST the time depends on weight decay alone, through the weight norm.

The works we cite each follow one quantity: the distance the kernel moves (Mohamadi et al., 2024), the weight norm (Khanh et al., 2026), the logit scale (Khanh, 2026), or the weight-decay rate (Kim, 2026; Pracher et al., 2026). This report separates the growth of the tangent kernel from its rotation, and compares on one grid the hypotheses that scale, rotation or the weight-decay rate sets the grokking time, and then asks the same question on a second task.

2 Related Work

Page budget: 1.2 pages.

Grokking. Power et al. (2022) report grokking on small algorithmic datasets. Nanda et al. (2023) reverse-engineer the algorithm a small transformer learns for modular addition and split training into memorisation, circuit formation and clean-up. Gromov (2023) gives closed-form weights that solve modular addition with a quadratic activation, and observes grokking under full-batch gradient descent with no explicit regularisation.

Lazy and rich training. In the infinite-width limit the tangent kernel stays constant, so the network trains as a kernel method (Jacot, Gabriel, and Hongler, 2020). Chizat, Oyallon, and Bach (2020) show that lazy training is a matter of scale. Multiplying a model's output

by a factor α brings its training closer to that of its linearisation as α grows. Kumar et al. (2024) use this factor to move a network between the two regimes, and propose that grokking is the transition from lazy to rich dynamics. Mohamadi et al. (2024) prove that a permutation-equivariant kernel method needs a constant fraction of the p^2 pairs to learn modular addition, and observe that the kernel moves far from its initial value only after the training data are fitted.

What sets the grokking time. Khanh et al. (2026) find that networks trained with weight decay grok when the weight norm reaches a critical value, and that clamping the norm above that value delays grokking exponentially. Khanh (2026) show that under cross-entropy the norm acts through the scale of the logits, and that under squared error it acts through a different route. Kim (2026, Cor. 5, Eq. 20, arXiv v1) derives, for linear models trained with heavy-ball momentum β and weight decay, a grokking time of about $(1-\beta)/(\eta\lambda)$ times a logarithm of an initial amplitude, and observes this scaling on modular addition. Pracher et al. (2026) predict, for homogeneous networks trained on the squared error with weight decay, that the grokking time is set by the product of the learning rate and the weight decay. Liu et al. (2023) describe the same relationship in terms of the weight norm. When weight decay is the only regulariser and the training loss is near zero after the data are fitted, a network starting with norm w_0 above a critical value w_c generalises after roughly $\ln(w_0/w_c)/(\eta\lambda)$ steps. They induce grokking on MNIST by increasing w_0.

The kernel under weight decay. For a network trained by gradient flow with L_2 strength λ, the kernel evolves as

$$\begin{aligned} \frac{d\Theta_t(x,x')}{dt} &= -2(k-1)\lambda\,\Theta_t(x,x') + T_t(x,x') + T_t(x',x), \\ T_t(x,x') &= -\sum_{x''} \partial_{\mu\nu} f_t(x)\,\partial_\mu f_t(x')\,\partial_\nu f_t(x'')\,\ell'(x''), \end{aligned} \tag{1}$$

where k is the degree of homogeneity of the network, the sum runs over the training inputs x'' and repeated parameter indices, and ℓ' is the derivative of the loss with respect to the output (Lewkowycz and Gur-Ari, 2021, Eqs. S5 and S6, arXiv v2). The weight decay appears only in the first term, which rescales the kernel. Only T_t can rotate it. At initialisation T_0 has expectation of order $1/N$ in the width N, a result that Lewkowycz and Gur-Ari (2021) take from Dyer and Gur-Ari (2020). In the infinite-width limit the kernel therefore only decays, with fixed eigenvectors (Lewkowycz and Gur-Ari, 2021, Thms. 1 and 2). **[TODO: Dyer and Gur-Ari (2020) is named here only as Lewkowycz and Gur-Ari's source. Add it to ref.bib only after checking it on arXiv.]**

Predictions. Table 1 sets out what each account predicts for the three tests of this report, on either task. If rotation sets the grokking time and T_t stays of order $1/N$ along training, a wider network rotates more slowly and groks later at a fixed weight decay. The weight decay should then matter little. A weight-decay clock (meaning that weight decay sets the grokking time) predicts the opposite. Each account was tested on its own, so these predictions have not been set against each other on one grid. The weight-decay accounts also leave open what sets the time without weight decay. Kim (2026, p. 3, arXiv v1) lists that case, and the creation of new directions by feature learning, among the things its mechanism does not explain. We believe that rotation of the tangent kernel measures one form of such feature learning.

Table 1: What each account predicts for the three tests. The exponents are those of $\log T = c + a \log N + b \log(\eta\lambda)$. "Not stated" means that the source makes no claim. Entries marked "ours" are our reading of the source. Tier 3 tests only the predictors that can be measured on a network, so it does not test $\eta\lambda$ alone (section 4.4).

Account	Fixed at grokking (Tier 1)	Width exponent a (Tier 2)	Decay exponent b (Tier 2)	Early predictor (Tier 3)
Rotation clock (our hypothesis)	R_t reaches a threshold	$a > 0$, since T_t shrinks with width (ours)	$b \approx 0$	R_t
Weight-decay clock (Kim, 2026; Liu et al., 2023; Pracher et al., 2026)	Not stated	$a \approx 0$; width can enter Kim's law only inside the logarithm (ours)	$b \approx -1$	$\eta\lambda$ alone
Norm or scale clock (Khanh, 2026; Khanh et al., 2026)	The weight norm reaches a critical value; in our terms S_t at grokking is fixed (ours)	Not stated	Not stated	S_t

3 Methods

Page budget: 1.8 pages.

3.1 Task, network and training

The task we train on is modular addition with modulus $p = 23$, at the baseline of Kumar et al. (2024, App. 8.3, arXiv v3). Each pair (a, b) of residues is encoded as the concatenation $x_{a,b} \in \mathbb{R}^{2p}$ of the one-hot codes of a and b. Its target $y_{a,b} \in \mathbb{R}^p$ is the one-hot code of $(a + b) \bmod p$. The dataset holds all $p^2 = 529$ pairs. Each seed draws its own split, with 476 pairs for training and 53 for testing, and its own initial weights.

The network is a two-layer perceptron without biases,

$$f(x;\theta) = N^{-1/2}\, W^{(2)} \phi\big(D^{-1/2} W^{(1)} x\big), \tag{2}$$

with input dimension $D = 2p$, width N, weights $W^{(1)} \in \mathbb{R}^{N \times D}$ and $W^{(2)} \in \mathbb{R}^{p \times N}$ drawn from $\mathcal{N}(0, 1)$, and the rectifier $\phi(h) = \max\{h, 0\}$. The factor $N^{-1/2}$ is the NTK parameterisation of Jacot, Gabriel, and Hongler (2020). Following Kumar et al. (2024), the trained predictor is $\tilde{f}(x;\theta) = \alpha[f(x;\theta) - f(x;\theta_0)]$ with learning rate $\eta = \eta_0/\alpha^2$. A larger α gives lazier training. The loss is the squared error averaged over the outputs and the training pairs. Training is full-batch gradient descent with weight decay, $\theta_{k+1} = (1 - \eta\lambda)\theta_k - \eta \nabla L(\theta_k)$. For plain gradient descent this is the same as L_2 regularisation (Kim, 2026). Table 10 (Appendix F) lists every setting.

3.2 The grid

Each axis of the grid tests one account. The output scale α sets how lazy the training is (Chizat, Oyallon, and Bach, 2020). The width N sets the size of T_t in (1), and so how fast the kernel can rotate. The weight decay $\eta\lambda$ sets the clock of Kim (2026). We call one combination of $(\alpha, N, \eta\lambda)$ a *cell*. The five runs of a cell share every setting and differ only in their seed, which

sets the split and the initial weights. The grid has $3 \times 6 \times 8 = 144$ cells and 720 runs, and every run finished.

3.3 Scale, rotation and alignment

We measure the tangent kernel of one scalar output, the sum of the logits divided by $\sqrt{p}$, $g(x;\theta) = p^{-1/2}\sum_c f_c(x;\theta)$. This is the pseudo-NTK of Mohamadi, Bae, and Sutherland (2022). On the 53 test pairs of a seed, the kernel at step t is the matrix with entries $[K_t]_{ij} = \langle \nabla_\theta g(x_i;\theta_t), \nabla_\theta g(x_j;\theta_t)\rangle$. The kernel of $\tilde{f}$ is α^2 times the kernel of f, so the ratios below do not depend on which of the two is used. With Y the one-hot targets of the test pairs and $C = I - \frac{1}{53}\mathbf{1}\mathbf{1}^\top$, we record

$$S_t = \frac{\|K_t\|_F}{\|K_0\|_F}, \qquad R_t = 1 - \frac{\langle K_t, K_0\rangle_F}{\|K_t\|_F\,\|K_0\|_F}, \qquad A_t = \frac{\langle CK_tC,\, CYY^\top C\rangle_F}{\|CK_tC\|_F\,\|CYY^\top C\|_F}. \tag{3}$$

The scale S_t is 1 at initialisation, and a value above 1 means the kernel grew. The rotation R_t is the kernel distance of Fort et al. (2020). It is 0 at initialisation and does not change when the kernel is rescaled. The alignment A_t is the centred alignment of Cortes, Mohri, and Rostamizadeh (2024) with the label kernel $YY^\top$, and it also does not change when the kernel is rescaled.

The usual measure of kernel movement is the relative variation $D_t = \|K_t - K_0\|_F / \|K_0\|_F$ (Geiger et al., 2020). It is a function of the other two,

$$D_t^2 = S_t^2 - 2S_t(1 - R_t) + 1, \tag{4}$$

(section A). A kernel that only grows has $R_t = 0$ and $D_t = |S_t - 1|$. A large D_t therefore does not show that the kernel rotated.

[TODO: Add a notation table here if the group wants one: S_t, R_t, A_t, T, t^*, a, b, with the value of each at initialisation and how to read it. It costs about a quarter of a page.]

3.4 Event times

The memorisation step of a run is the first checkpoint at which its training accuracy reaches 0.99. Its grokking step T is the first checkpoint at which its test accuracy reaches 0.95. Appendix E repeats the analysis at the levels 0.80 and 0.90. Both events are counted from step 0. A run that does not reach a level within 200 000 steps is kept, and its grokking step is recorded as censored at the budget (200 000 steps). The events are read on the checkpoint grid described above, so each one is known to lie between two checkpoints. The gap to the checkpoint before is at most 9.7% of the grokking step and 10.9% of the memorisation step.

3.5 Analysis

Thresholds (Tier 1). For each statistic we take its value at the grokking step of every run that groks, and measure the spread across runs by the standard deviation (SD). A statistic can have a small spread because it stays in a narrow band all through training, so we include a placebo test to separate the two cases. The placebo test reads each run at the grokking step of another run, drawn by a random permutation with no fixed points, and measures the SD again. The tightening ratio is the median placebo SD over 2 000 permutations divided by the SD at the true events, and p is the share of permutations whose SD is no larger. A ratio near 1 means the statistic is not tied to the event. The SD carries the units of its statistic, so the SDs of different statistics cannot be compared directly.

Scaling (Tier 2). We fit the log-normal accelerated failure time model

$$\log T = c + a \log N + b \log(\eta\lambda) + \gamma \log N \log(\eta\lambda) + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1), \tag{5}$$

[TODO: i think this needs more explanation for our audience] by maximum likelihood. A censored run contributes the probability that its grokking step exceeds the budget (Kalbfleisch and Prentice, 2002). Both logarithms are centred at the middle of the grid, so a and b are the slopes there. The fits use lifelines (Davidson-Pilon, 2019). The intervals resample whole cells and then seeds within cells, so that they include the misfit of the cells about the fitted plane.

Prediction (Tier 3). We read S_t, $\log R_t$ and A_t at $t^* = 2\,430$, a saved step before any run reaches the grokking level, and ask how well they predict the grokking step. The five seeds of a cell are close copies, so whole groups of runs are always held out. Two tests are used. The first holds out one cell at a time, which tests interpolation. Kernel ridge regression with a Gaussian kernel is fitted on each subset of the statistics, which shows which statistics carry the information. The second holds out one value of α at a time, which tests extrapolation to an unseen laziness. It compares a linear accelerated failure time model on each statistic alone, gradient-boosted trees with the accelerated failure time objective (Chen and Guestrin, 2016) on all three, and a gated recurrent unit (Cho et al., 2014) that reads all three at the 25 saved steps up to t^*. The weight decay does not enter as a predictor, since it is a setting and not a measurement (section 4.4). As a reference, the second test also fits the trees to the settings N and $\eta\lambda$ together and to each one alone. Two scores are used. R^2 is the share of the spread in $\log T$ explained on held-out runs. The concordance index is the share of pairs of runs that a model puts in the right order, so that 0.5 is a coin toss (Harrell et al., 1982). Appendix D gives the settings of each model.

3.6 The second task

In tier 4 we apply the same questions from Tiers 1 and 2 to the MNIST dataset using the same setup with which Liu et al. (2023) induce grokking. The network is trained on a subset of 1000 images and the initial weights are multiplied by a scale $s > 1$ so the weight decay must shrink the norm before the network generalises. This differs from section 3.1 which subtracts $f(x;\theta_0)$ so that every run is centered and starts from the zero function. Centring eliminates grokking delay on MNIST and instead the train and test accuracies rise at relatively the same time which is why it cannot be used. The architecture is a bias-free three-layer rectifier network of width N in the standard parameterisation, trained by minibatch gradient descent without momentum, with the learning rate reset from the curvature every 10000 steps. The values tested on the grid are $s \in \{3, 5, 7, 9\}$, $N \in \{100, 200, 400, 800\}$ and seven decays from 0 to 4×10^{-5}, each with 5 seeds. Each run has a budget of 500000 steps and the grokking level is 0.80 since no run reaches 0.95 (Appendix G).

4 Experiments

4.1 Baseline: the reproduction of Kumar et al.

Page budget: 0.8 page.

Finding. Our runs reproduce the baseline of Kumar et al. (2024). The network memorises within a few thousand steps and groks much later, and a lazier network groks later. Over this time the kernel grows several-fold while R_t stays below 0.09.

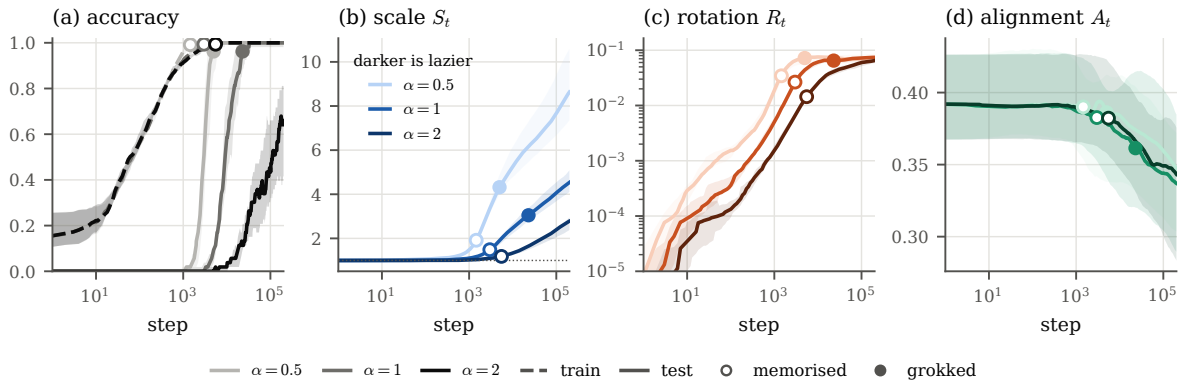


Figure 1: Training at the baseline of Kumar et al. (2024) ($N = 100$, $\eta\lambda = 0$) for three values of α. Each line is the median over 5 seeds and the shading is their range. Darker lines are lazier networks. A hollow marker is the median memorisation step and a filled marker the median grokking step. No seed groks at $\alpha = 2$. R_t is on a log scale.

At $\alpha = 0.5$ and $\alpha = 1$ every seed memorises and then groks, with median grokking steps of 4 981 and 23 000 (Figure 1). At $\alpha = 2$ no seed groks within 200 000 steps. These grokking steps and their order in α are close to those of Kumar et al. (2024, Fig. 2(c), arXiv v3). Appendix B compares the settings and the event steps one by one.

By (4) the squared variation splits into a scale part $(S_t - 1)^2$ and a rotation part $2S_tR_t$. At $\alpha = 1$ the scale part is 90% of D_t^2 at grokking. The distance D_t thus mostly records the growth of the kernel, which is why we measure S_t and R_t separately.

4.2 Tier 1: is there a threshold?

Page budget: 1.6 pages.

Finding. At a fixed width, rotation is the statistic most tied to the grokking step, but reaching that rotation does not make a network grok. The alignment at grokking barely differs from its value at initialisation. Across widths, no statistic takes a fixed value at grokking.

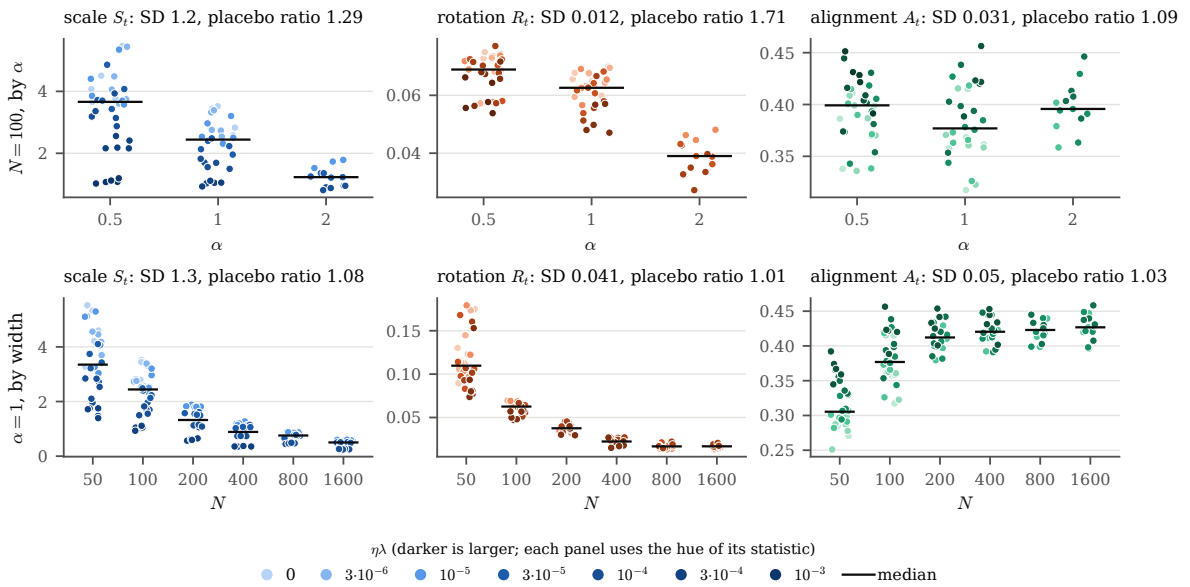


Figure 2: The kernel statistics at the grokking step of every run that groks. Top row: $N = 100$, grouped by α. Bottom row: $\alpha = 1$, grouped by width. Colour is $\eta\lambda$, darker for larger, and bars are medians. Each title gives the SD at grokking and the tightening ratio of the placebo test (Table 2).

Table 2: Spread of each statistic across runs that grok, as a standard deviation, at the grokking step, at the memorisation step and at another run's grokking step (placebo, median over 2 000 permutations). The ratio is the placebo SD over the SD at grokking, and p is the share of permutations with an SD no larger than the true one.

Runs	Statistic	At grokking	At memorisation	Placebo	Ratio	p
$N = 100$ (78 runs)	S_t	1.2	0.42	1.6	1.29	< 0.0005
	R_t	0.012	0.0094	0.02	1.71	< 0.0005
	A_t	0.031	0.02	0.034	1.09	0.029
$\alpha = 1$ (130 runs)	S_t	1.3	0.46	1.4	1.08	0.258
	R_t	0.041	0.035	0.041	1.01	0.423
	A_t	0.05	0.033	0.052	1.03	0.079

At $N = 100$, R_t is the statistic most tied to the event. Its SD widens 1.71-fold when each run is read at another run's grokking step, against 1.29-fold for S_t and 1.09-fold for A_t (Table 2). No permutation gave R_t or S_t a spread as small as the true one. The alignment tightens only a little ($p = 0.029$). Its median over the runs that grok is 0.398 at initialisation and 0.406 at grokking. Across widths no statistic passes the placebo test at level 0.95. At $\alpha = 1$ the ratios are 1.08, 1.01 and 1.03, and no p is below 0.05. At grokking R_t falls from 0.108 at $N = 50$ to 0.019 at $N = 1600$, and S_t falls from 4.11 to 0.58 (medians over all α and $\eta\lambda$). At level 0.80 (section E) the scale and the alignment tighten slightly across widths, and at 0.90 only the alignment does. R_t does not at any level. Pooled over all 355 runs that grok, S_t tightens 1.69-fold ($p < 0.0005$). The grokking steps of that pool span nearly two decades while S_t keeps growing, so this may reflect growth more than a threshold.

Reaching the rotation at which runs grok is not enough to grok. There are 34 cells in which some seed memorises but no seed groks. In 8 of them, at least one seed reaches the median R_t

at grokking of its width before the budget runs out. One of them is the baseline cell at $\alpha = 2$ (section 4.1).

Wider networks grok with less rotation but take longer, which suggests that width sets the rate at which the kernel rotates. Section 4.3 tests this. Our proposal expected the alignment to cross a threshold shared by every cell, and that does not hold. Other claims of a fixed value at grokking, such as the critical weight norm of Khanh et al. (2026), can be given the same placebo check.

4.3 Tier 2: width or weight decay?

Page budget: 2.0 pages.

Finding. The grokking time depends on both width and weight decay, so neither the rotation clock nor the weight-decay clock of Table 1 holds on its own. Width sets the rotation of the kernel. At a fixed width, weight decay shrinks its scale.

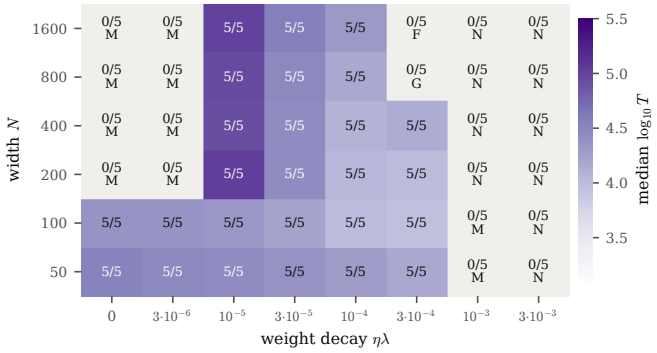


Figure 3: Grokking time and outcome over width and weight decay at $\alpha = 1$. Colour is the median $\log_{10} T$ over the seeds that grok, shown when at least 3 of 5 grok. Each square is one cell and gives the seeds that grok out of 5. A cell in which none grok also shows the most common phase at the last step: M memorises only, F forgets, N never fits, and G ends above test accuracy 0.9 without reaching 0.95.

Without weight decay, only $N = 50$ and $N = 100$ grok within the budget (Figure 3). The wider networks memorise and stop there. From $\eta\lambda = 10^{-5}$ to 10^{-4} every seed at every width groks, and the time falls with $\eta\lambda$ at every width. The largest weight decay at which every seed groks is 3×10^{-4} for $N \leq 400$ and 10^{-4} for $N = 800$ and $N = 1600$. At 10^{-3} and above no width groks. A wider network thus groks later at the same weight decay and stops grokking at a smaller one, so the weight decay that speeds grokking depends on the width.

[TODO: The figure of the grokking time against weight decay at each width (E5, `figures/E5_scaling.pdf`) is commented out here, because Table 3 gives the same result and the figure is hard to read without an explanation. Consider it for the seminar, where we can explain how to read it.]

Table 3: Fits of (5) at $\alpha = 1$ beside the predictions of Table 1. Brackets give 95% intervals from a bootstrap that resamples whole cells and then seeds within cells. The main row uses the 90 runs with $10^{-5} \le \eta\lambda \le 10^{-4}$, where every seed at every width groks. The other rows are checks.

Fit	a	b	γ	σ
Rotation clock predicts	> 0	≈ 0		
Weight-decay clock predicts	≈ 0	≈ -1		
Main range, level 0.95	$+0.24$ [$+0.08$, $+0.40$]	-0.63 [-0.92, -0.46]	-0.16 [-0.37, $+0.05$]	0.34
Adding $\eta\lambda = 3 \times 10^{-4}$	$+0.36$ [$+0.03$, $+0.62$]	-0.21 [-0.60, $+0.14$]	$+0.14$ [-0.24, $+0.48$]	0.80
Level 0.80	$+0.30$ [$+0.17$, $+0.45$]	-0.55 [-0.74, -0.39]	-0.17 [-0.36, -0.02]	0.29
Level 0.90	$+0.27$ [$+0.13$, $+0.41$]	-0.60 [-0.84, -0.42]	-0.16 [-0.37, $+0.03$]	0.31
Memorisation step	-1.53 [-1.80, -1.32]	$+0.07$ [-0.13, $+0.31$]	$+0.05$ [-0.23, $+0.29$]	0.38

Both exponents are nonzero, and both intervals exclude zero ($a = +0.24$, $b = -0.63$). A wider network groks later, as the rotation clock predicts, and more weight decay shortens the time, as the weight-decay clock predicts. Neither exponent takes the value its account predicts. Wider networks memorise sooner, so the width effect on grokking is not a delay in fitting the training data (memorisation row). The estimate of γ is negative, which would mean that weight decay shortens the time more at large width. Its interval includes zero, so the data do not fix its sign. Adding the cells at $\eta\lambda = 3 \times 10^{-4}$, where the widest networks stop grokking, changes the fit: b moves to -0.21 and its interval includes zero. The exponents therefore hold for the range in which every seed groks.

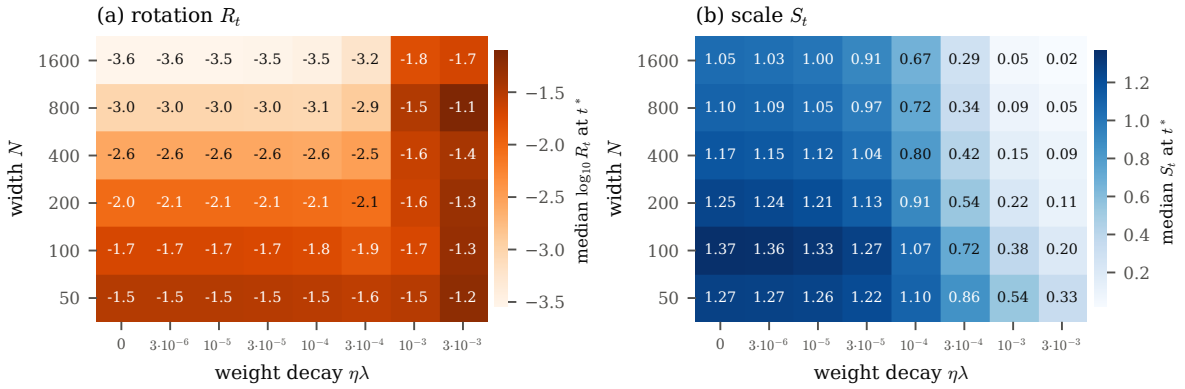


Figure 4: Width sets the rotation of the kernel, and weight decay shrinks its scale. Medians over seeds at $\alpha = 1$, read at $t^* = 2\,430$, before any run groks: (a) $\log_{10} R_t$ and (b) S_t.

Up to $\eta\lambda = 3 \times 10^{-4}$, the rotation at t^* depends on width and hardly on weight decay (Figure 4). Narrow networks rotate the most. The scale falls with weight decay at every width. Among the runs that grok, width alone explains 0.66 of the spread of $\log R_t$, and weight decay 0.18. The spread of S_t is shared: α explains 0.28, width 0.22 and weight decay 0.26. This links Tier 2 back to Tier 1. Width sets the rotation, weight decay is one of the settings that set the scale, and the grokking time depends on both.

4.4 Tier 3: predicting the grokking time from the kernel

Page budget: 1.6 pages.

Finding. Read before any run groks, rotation is the best single predictor of the grokking time on held-out cells. Scale adds information that rotation lacks, and alignment adds little. On an unseen α, the three statistics together rank runs better than any one of them alone. The settings N and $\eta\lambda$ rank them better still, and better than either setting alone, which supports the mixed clock of section 4.3.

Each account in Table 1 names a different early predictor: R_t for the rotation clock, S_t for the norm or scale clock, and $\eta\lambda$ alone for the weight-decay clock. This tier tests the first two, with A_t beside them. It does not test $\eta\lambda$, which we set and know exactly, while a general network reports neither its effective regularisation nor its laziness. Its tangent kernel can be measured. We want predictors that carry over to real models, so this tier asks how much of the grokking time the kernel statistics recover on their own. A model given the settings N and $\eta\lambda$ serves as a reference that knows more than a general network would report.

Table 4: Kernel ridge regression on subsets of the statistics at t^*, for the 355 runs that grok, scored with each of the 74 cells held out in turn. The last column is the share of the spread of each quantity that lies between cells.

Statistics	R^2	Concordance	Quantity	Between cells
A_t	0.05	0.55	$\log T$	0.96
S_t	0.27	0.67	S_t	0.99
$\log R_t$	0.35	0.70	$\log R_t$	0.99
S_t and $\log R_t$	0.51	0.76	A_t	0.56
$\log R_t$ and A_t	0.39	0.72		
S_t and A_t	0.23	0.65		
All three	0.55	0.78		

Taken alone, $\log R_t$ explains the most of the grokking time on held-out cells (Table 4). Adding S_t to $\log R_t$ raises R^2 from 0.35 to 0.51, so rotation does not carry all of the information. Adding A_t to both raises it only to 0.55. Between 0.96 and 0.99 of the spread of $\log T$, S_t and $\log R_t$ lies between cells (Table 4). The statistics are therefore almost fully set by the cell, and they say nothing about which seed of a cell groks first.

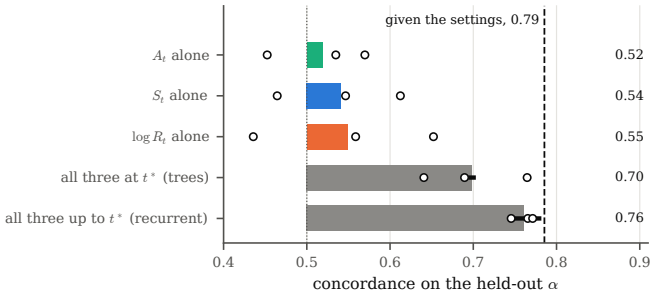


Figure 5: Predicting the grokking step of runs with an unseen α. Each model is fitted on two values of α and scored on the third by the concordance index with censoring. Each statistic alone enters a linear accelerated failure time model. The trees read all three statistics at t^*, and the recurrent network reads them at the 25 saved steps up to t^*. Bars are the mean over the three held-out values and dots each value. Black lines give the range over seeds (5 for the trees, 3 for the recurrent network). The dashed line is the trees given the settings N and $\eta\lambda$.

On an unseen α each statistic alone ranks runs only a little better than a coin toss (Figure 5):

0.52 for A_t, 0.54 for S_t and 0.55 for $\log R_t$. Gradient-boosted trees on all three reach 0.70, and the recurrent network that reads their history up to t^* reaches 0.76. The reference model given the settings reaches 0.79, more than any model of the kernel statistics. Nanda et al. (2023, p. 1, arXiv v3) call a progress measure a metric that varies smoothly, precedes the phase transition and is causally linked to it. Rotation meets the first two conditions here. We have not tested a causal link, and reaching a rotation does not make a network grok (section 4.2). Rotation also gives no earlier or better warning than the settings themselves.

The settings reference combines width and weight decay. The same trees given $\log N$ alone reach 0.67, and given $\eta\lambda$ alone they reach 0.72 (0.71 to 0.73 over seeds). Each setting alone therefore ranks runs worse than the two together. This supports the mixed clock of section 4.3, in which the grokking time depends on both width and weight decay.

4.5 Tier 4: a second task, MNIST digits

Page budget: 1.0 page.

Finding. On MNIST the weight-decay clock of Table 1 holds exactly. The grokking time falls inversely proportional to weight decay and does not depend on width. At grokking, the weight norm reaches a fixed value, and the kernel shrinks with it while its shape remains nearly unchanged.

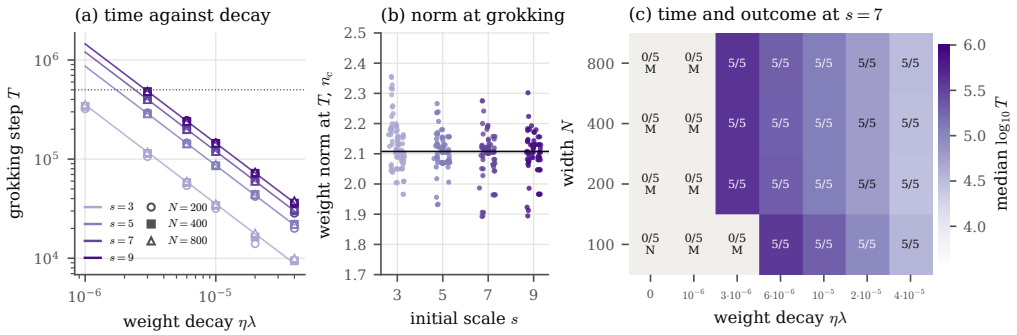


Figure 6: MNIST at $s = 7$. Panels (a) and (b) use widths 200 to 800; (c) adds width 100, which reaches the level later and off the law (Appendix G). (a) The grokking step against weight decay, medians over 5 seeds, coloured by the initial scale s with one marker per width. The lines are $T = \ln(s/n_c)/(\eta\lambda)$ with the mean n_c of panel (b), and the dotted line is the budget. (b) The weight norm at grokking in units of the norm at $s = 1$, one point per run, with its mean. (c) The map of Figure 3 at $s = 7$: the median grokking step over the seeds where at least 3 of 5 grok, the seeds that grok out of 5, and the phase letter of Figure 3 where none does.

In the main grid range, $s = 7$ at $N \in \{200, 400, 800\}$ with $\eta\lambda$ from 3×10^{-6} to 10^{-5}, every seed groks, and the fit of (5) gives $a = -0.02$ $[-0.07, +0.02]$ and $b = -1.00$ $[-1.05, -0.96]$, against $+0.24$ and -0.63 on modular addition (Table 3), with a residual σ of 0.03 against 0.34.

This is likely a direct result of the weight norm. The product $\eta\lambda T$ is constant within each scale and increases with s (Figure 6a), and at grokking the norm is 2.11 times the norm at $s = 1$, with a standard deviation of 0.07 (3%) over the 314 runs that reach the level, across every scale, width and decay (Figure 6b). As in Liu et al. (2023) the norm follows pure decay and the time is given by the law $T = \ln(s/n_c)/(\eta\lambda)$ (Appendix G). As the norm falls, the kernel shrinks too where at grokking, it is between 0.0012 and 0.0050 of its initial value. As in modular addition the rotation decreases with width but in MNIST falls sharply going from 0.045 at $N = 200$ to 0.002 at $N = 800$ while T remains approximately the same. The Tier 1 placebo test cannot

distinguish these statistics here (Appendix G).

The two tasks therefore grok by different routes. Grokking in MNIST is set by the Liu et al. (2023) problem construction where the enlarged initial function must decay and the kernel need only shrink before generalisation occurs. On modular addition the delay happens because of the centring, and Mohamadi et al. (2024) prove that a fixed permutation-equivariant kernel cannot learn the task from a constant fraction of the pairs. The kernel must change shape first and does so at a rate that depends on width.

5 Limitations

Page budget: 0.6 page.

Every modular addition run uses the NTK parameterisation, one task and one architecture. A width effect measured here belongs to this parameterisation. Each cell has 5 seeds. The events are known only to the resolution of the checkpoint grid. The fits in Tier 2 depend on which cells enter, and the decay edge bends the plane (Table 3). The kernel statistics say nothing about which seed of a cell groks first, and the reference model given the settings predicts better than any kernel model. The alignment here uses the kernel of the summed logits. The tangent kernel of all p logits may behave differently. The centred predictor is not homogeneous, so the theorem behind (1) applies to our network only as a guide. The proposal also planned a clamped-norm arm, a sweep over p and the training fraction, and a cross-entropy arm. They were not run. The proposal defined the events by the loss. The report uses accuracy (section 3.4).

The MNIST runs differ in not only the task but the architecture and method for inducing grokking, with the most impactful of these being the uncentred predictor. Our comparison assumes that none of these changes which clock applies. With only two tasks to compare we cannot tell which of them, if any, is responsible for the different result. Tier 4 tests which clock sets the time once a delay exists, not whether rotation can cause one.

6 Conclusion

Page budget: 0.3 page.

We asked what sets the grokking time: the shape of the tangent kernel, its scale, or the weight-decay rate. On modular addition, rotation matters most and is not enough on its own. At a fixed width it is the statistic most tied to the grokking step, yet networks can reach that rotation without grokking. The grokking time grows with width and falls with weight decay. Width sets the rotation, and at a fixed width weight decay shrinks the scale. Early rotation and scale together predict the grokking time better than either one alone, and width and weight decay together predict it better than either setting alone. On MNIST, Liu et al. (2023) create a delay by increasing the initial function's scale. The weight-decay clock predicts the grokking time exactly using the law $\ln(s/n_c)/(\eta\lambda)$, with no dependence on width. The kernel shrinks but rotates very little where width affects only the rotation. Weight decay therefore sets the grokking time through the weight norm on both tasks, and width enters only on modular addition, where a fixed kernel cannot solve the task and the kernel must change shape first. What is new is the split into scale and rotation, the placebo test, the crossed grid on two tasks, and prediction from quantities that any network exposes. A next step is to test whether the early rate of rotation falls as $1/N$, and whether a kernel frozen at each step can generalise on either task.

Contribution

[TODO: One line for each member, from the group.]

- Al-Saffi: **[TODO:]**
- Chong: **[TODO:]**
- D'Mello: **[TODO:]**
- Kienzle: **[TODO:]**
- Rieger: **[TODO:]**

Acknowledgement

[TODO: Anyone outside the group who advised on the project.]

Use of AI tools

[TODO: The task sheet asks each deliverable to state how AI tools were used. The group should agree on this text. A draft follows.] The group used an AI assistant (Claude, by Anthropic) to draft parts of the code, to restructure this report, and to check references against their arXiv pages. Group members checked every result and every reference before submission.

A The identity for the variation

Page budget: 0.5 page (appendices: 5 pages in all).

Expanding the squared norm gives $\|K_t - K_0\|_F^2 = \|K_t\|_F^2 - 2\langle K_t, K_0\rangle_F + \|K_0\|_F^2$. Dividing by $\|K_0\|_F^2$ and using $\langle K_t, K_0\rangle_F = (1-R_t)\|K_t\|_F\|K_0\|_F$ from (3) gives $D_t^2 = S_t^2 - 2S_t(1-R_t)+1$. By the Cauchy-Schwarz inequality, $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$, and then $D_t = |S_t - 1|$. **[TODO: Check against the golden report, Appendix A.1.]**

B The baseline against Kumar et al.

Page budget: 0.75 page.

[TODO: Decide whether the report needs this appendix for the reproduction. Section 4.1 states the result, and this appendix holds the check setting by setting, the event steps and the full split of D_t^2. If the appendix goes, keep one sentence in section 4.1 saying that our networks memorise several times sooner than theirs. CLAUDE.md (Rule 3) asks the report to state where our results and a source disagree.]

Table 5 sets each setting of our baseline beside Kumar et al. (2024, arXiv v3). We read their Appendices 8.1 and 8.3, Section 3 and Figure 2. They do not state the parameterisation or the initialisation. Table 6 sets our event steps beside the steps read by eye from their Figure 2, which shows one run for each α. Their grokking at $\alpha = 0.5$ and $\alpha = 1$ falls close to ours. Their training accuracy rises much more slowly. At $\alpha = 1$ it reaches 100% near 20 000 epochs, while our runs reach 0.99 by about 3 000 steps. The parameterisation and the initialisation they leave unstated may explain this gap. The reproduction therefore matches on the grokking time and on the order in α, and does not match on the memorisation time. They also sweep $\alpha = 1.5$, which our grid leaves out.

Section 4.1 gives the split of the variation at grokking for $\alpha = 1$. The other events and values of α follow. At $\alpha = 1$ the scale part $(S_t - 1)^2$ is 76% of D_t^2 at memorisation, 90% at grokking and 95% at the last step. At the last step S_t is 4.5, while R_t is only 0.074. The scale part is 94% at grokking for $\alpha = 0.5$. For $\alpha = 2$ it is 48% at memorisation and 90% at the last step.

Table 5: The baseline setting by setting. "Where" is the place in Kumar et al. (2024, arXiv v3) that gives the setting.

Setting	Kumar et al. (2024)	This report	Where
Task	$a + b \bmod p$, $p = 23$	$a + b \bmod p$, $p = 23$	App. 8.3
Encoding	two one-hot codes, $D = 2p$	two one-hot codes, $D = 2p$	App. 8.3
Output	p outputs, one-hot target	p outputs, one-hot target	App. 8.3
Training pairs	90% of p^2	90% of p^2 (476 of 529)	App. 8.3
Width	$N = 100$	$N = 100$ (grid: 50 to 1600)	App. 8.3
Network	one hidden layer	one hidden layer, no biases, rectifier	App. 8.3
Parameterisation	not stated	NTK, weights from $\mathcal{N}(0, 1)$	
Loss	mean squared error	mean squared error	Sec. 3
Optimiser	vanilla full-batch GD, no weight decay	full-batch GD; $\eta\lambda = 0$ at the baseline	App. 8.3
Predictor	$\alpha[f(x;\theta) - f(x;\theta_0)]$	the same	App. 8.1
Learning rate	$\eta = 100$; $\eta = \eta_0/\alpha^2$	$\eta = 100/\alpha^2$	App. 8.1, 8.3
α	$\{0.5, 1, 1.5, 2\}$	$\{0.5, 1, 2\}$	Fig. 2(c)
Budget	100 000 epochs (Fig. 2(c))	200 000 steps	Fig. 2(c)
Seeds	one run shown	5 per cell	Fig. 2

Table 6: Event steps at $N = 100$ and $\eta\lambda = 0$. The values of Kumar et al. (2024) are read by eye from their Fig. 2(a) and (c), so they are approximate.

	Kumar et al. (2024), read from Fig. 2		This report, median [range] over 5 seeds	
α	Train 100%	Test 100%	Memorisation	Grokking (0.95)
0.5	≈4 000	≈5 000	1 455 [1 313, 1 612]	4 981 [3 662, 6 776], 5/5
1	≈20 000 to 25 000	≈25 000 to 27 000	2 983 [2 692, 2 983]	23 000 [15 395, 34 000], 5/5
1.5	≈65 000	≈70 000	not in the grid	
2	≈90 000	about 25% at 100 000	5 520 [4 496, 6 000]	0/5 by 200 000

C Outcomes over the whole grid

Page budget: 1.0 page.

Table 7 gives the outcome of every cell. An entry gives the seeds that grok at level 0.95 out of 5. After a colon it gives the median grokking step in thousands, when at least 3 seeds grok. A cell in which no seed groks shows 0 and the most common phase at the last step, as in Figure 3. Every seed groks in 35 of the 48 cells at $\alpha = 0.5$, 26 at $\alpha = 1$ and 8 at $\alpha = 2$.

Table 7: Outcomes over the whole grid at level 0.95: seeds that grok out of 5, and the median grokking step in thousands.

α	N	0	3×10^{-6}	10^{-5}	3×10^{-5}	10^{-4}	3×10^{-4}	10^{-3}	3×10^{-3}
0.5	50	5: 7	5: 7	5: 8	5: 7	5: 7	5: 5	5: 6	0 N
	100	5: 5	5: 5	5: 4	5: 4	5: 4	5: 4	5: 4	0 N
	200	3: 27	5: 26	5: 14	5: 7	5: 5	5: 4	5: 6	0 N
	400	0 M	5: 70	5: 19	5: 9	5: 6	5: 4	0 M	0 N
	800	0 M	5: 106	5: 33	5: 12	5: 7	5: 6	0 N	0 N
	1600	0 M	5: 157	5: 48	5: 17	5: 8	5: 9	0 N	0 N
1	50	5: 33	5: 30	5: 27	5: 24	5: 19	5: 15	0 M	0 N
	100	5: 23	5: 23	5: 21	5: 17	5: 10	5: 9	0 M	0 N
	200	0 M	0 M	5: 106	5: 27	5: 11	5: 10	0 N	0 N
	400	0 M	0 M	5: 82	5: 27	5: 12	5: 14	0 N	0 N
	800	0 M	0 M	5: 88	5: 30	5: 15	0 G	0 N	0 N
	1600	0 M	0 M	5: 99	5: 37	5: 20	0 F	0 N	0 N
2	50	0 M	0 M	1	2	1	0 M	0 N	0 N
	100	0 M	0 M	3: 95	5: 69	5: 43	0 M	0 N	0 N
	200	0 M	0 M	0 M	5: 86	5: 40	0 M	0 N	0 N
	400	0 M	0 M	0 M	5: 83	5: 60	0 N	0 N	0 N
	800	0 M	0 M	0 M	5: 94	0 G	0 N	0 N	0 N
	1600	0 M	0 M	0 M	5: 98	0 M	0 N	0 N	0 N

D Settings of the prediction models

Page budget: 1.0 page.

All models use the 630 runs with $\eta\lambda \leq 10^{-3}$, with S_t, $\log R_t$ and A_t read at t^*. The grokking step of a censored run is known only to exceed 200 000. Whole groups are held out: one value of α at a time (Figure 5) or one cell at a time (Table 4). The file `report_v2/code/models.py` fixes every setting below.

Linear models. The log-normal accelerated failure time model of a single statistic standardises it on the training fold and uses a ridge penalty of 0.01.

Kernel ridge. The kernel is Gaussian on standardised inputs. Its penalty in $\{0.01, 0.1, 1\}$ and its width in $\{0.1, 0.5, 2\}$ are chosen by grouped 5-fold cross-validation over the training cells, inside every outer fold. It is fitted to $\log T$ on the training runs that grok.

Gradient-boosted trees. The trees use the accelerated failure time objective with normal errors of scale 1 and a subsample of 0.8. The search covers depths 1 to 4, learning rates 0.03 and 0.1, and minimum child weights 1, 5 and 10, with up to 1 000 rounds and early stopping after 50. The setting with the best concordance on 20% of the training cells is refitted on all training runs. The seeds are 0 to 4. The reference model is the same search on $\log N$ and $\log_{10}(\eta\lambda)$, with $\eta\lambda = 0$ set to 10^{-6}. Its two parts are the same search on $\log N$ alone and on $\log_{10}(\eta\lambda)$ alone.

Recurrent network. One gated recurrent layer with 16 units reads A_t, S_t and R_t at the 25 saved steps up to t^*. R_t enters without a logarithm because it is 0 at step 0. The inputs are standardised on the training fold. A linear head gives the location and a learned parameter the scale of a log-normal accelerated failure time loss. Training uses Adam with learning rate 0.01

and weight decay 10^{-4}, on the full batch, for up to 800 epochs. Every 10 epochs the loss on 20% of the training cells is checked, and the best weights are kept. Early stopping chose between 70 and 350 epochs. The seeds are 0 to 2.

E Sensitivity to the grokking level

Page budget: 0.5 page.

Table 8 repeats the placebo test at the levels 0.80 and 0.90, and Table 9 repeats the main fit. At $N = 100$ the order of the statistics is the same at every level. R_t tightens most, with ratios of 1.83 at 0.80 and 1.79 at 0.90, and S_t comes next. Across widths R_t never passes. At 0.80 the scale and the alignment tighten slightly across widths, with p of 0.024 and 0.001 and ratios of only 1.25 and 1.08. At 0.90 only the alignment does ($p = 0.013$); the scale has $p = 0.085$. The exponents a and b keep their signs at every level, and both intervals exclude zero.

Table 8: The placebo test of Table 2 at three grokking levels. The number of runs that grok is in brackets.

Level	Runs	Statistic	SD at grokking	Placebo SD	Ratio	p
0.80	$N = 100$ (86)	S_t	1.1	1.59	1.45	0.000
		R_t	0.0122	0.0223	1.83	0.000
		A_t	0.0291	0.033	1.13	0.002
0.80	$\alpha = 1$ (136)	S_t	1.01	1.26	1.25	0.024
		R_t	0.0376	0.0391	1.04	0.131
		A_t	0.0446	0.048	1.08	0.001
0.90	$N = 100$ (79)	S_t	1.18	1.58	1.34	0.000
		R_t	0.0118	0.021	1.79	0.000
		A_t	0.0307	0.0335	1.09	0.029
0.90	$\alpha = 1$ (134)	S_t	1.15	1.35	1.17	0.085
		R_t	0.0392	0.0403	1.03	0.195
		A_t	0.0475	0.0501	1.06	0.013
0.95	$N = 100$ (78)	S_t	1.25	1.6	1.29	0.000
		R_t	0.0119	0.0203	1.71	0.000
		A_t	0.0313	0.0341	1.09	0.029
0.95	$\alpha = 1$ (130)	S_t	1.33	1.44	1.08	0.258
		R_t	0.0408	0.0411	1.01	0.423
		A_t	0.0504	0.0521	1.03	0.079

Table 9: The main fit of Table 3 at three grokking levels, with intervals that resample cells and seeds.

Level	a	b	γ	σ
0.80	+0.30 [+0.17, +0.45]	−0.55 [−0.74, −0.39]	−0.17 [−0.36, −0.02]	0.29
0.90	+0.27 [+0.13, +0.41]	−0.60 [−0.84, −0.42]	−0.16 [−0.37, +0.03]	0.31
0.95	+0.24 [+0.08, +0.40]	−0.63 [−0.92, −0.46]	−0.16 [−0.37, +0.05]	0.34

The intervals in Table 3 resample cells as well as seeds. The misfit of the cells about the fitted plane has an SD of 0.32 in $\log T$, against 0.11 for the spread between the seeds of one cell. An interval that resamples seeds alone holds the cells fixed and is much narrower: for a it runs from +0.21 to +0.26.

F Settings of the modular addition runs

Table 10: Every setting of the runs. One combination of $(\alpha, N, \eta\lambda)$ is a cell (section 3.2), and each cell has 5 seeds.

Setting	Value
Modulus, pairs	$p = 23$, all 529 pairs; 476 for training and 53 for testing
Split and seeds	Model seed $s \in \{0, \ldots, 4\}$; the split uses data seed $42 + s$
Network	Equation. (2), no biases, rectifier, weights from $\mathcal{N}(0, 1)$
Width N	$\{50, 100, 200, 400, 800, 1600\}$
Output scale α	$\{0.5, 1, 2\}$
Learning rate	$\eta = \eta_0/\alpha^2$ with $\eta_0 = 100$
Weight decay $\eta\lambda$	$\{0, 3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$
Optimiser and loss	Full-batch gradient descent, no momentum; mean squared error
Budget	200 000 steps
Checkpoints	307 per run: step 0, 120 log-spaced steps to 200 000, and every 1 000 steps
Saved weights	48 steps per run
Runs	720, all finished
Compute	18.9 h summed over runs; 2.7 h elapsed on 12 workers of one Mac

G Settings of the MNIST runs

Page budget: 0.75 page.

Table 11 gives every setting of the MNIST sweep beside the modular addition setting it replaces. Table 12 gives the fit of section 4.5 with its check rows, and the exponents are the same in every row. The runs of one width and one scale were trained together as one fused model on one H100, in 134 summed GPU hours, with TF32 matrix multiplies. A second sweep in full float32 gave the same grokking outcome in every one of the 560 runs and the same memorisation outcome in all but 3.

Some outcomes stay out of section 4.5. Width 100 reaches the level in 87 of its 120 runs with decay, but its test accuracy at memorisation is already 0.72, so there is little delay to time; it reaches the level later than the wider networks at the same decay, with a lower and scattered norm, so it does not follow the law and is left out of the fits. We read this as a limit of capacity at 1 000 images. Without weight decay no run reaches the level within the budget. The three widths of the main range memorise within a few thousand steps of each other, and a network at $s = 3$ already has test accuracy 0.52 when it memorises. The norm follows pure decay, $\|\theta_T\| = \|\theta_0\| e^{-\eta\lambda T}$, to within 0.008 of the initial norm, and the law of section 4.5 has correlation 0.9963 with the measured product $\eta\lambda T$. The alignment at grokking is 0.396 against 0.421 at initialisation, as on modular addition. Under the placebo every statistic tightens, because each is a function of the step and the grokking steps span more than a decade: the ratios are 1.29 to 8.06 at $N = 200$ and 1.30 to 47.54 across widths. Tier 3 is not repeated: with σ of 0.03, an early read of the norm predicts the time and the rotation does not. The centred pilot of section 3.6 used the two-layer network of section 3.1 on MNIST with initial weights times 3, full-batch gradient descent, $\eta\lambda = 10^{-5}$, one seed and 200 000 steps, at $\alpha \in \{1, 0.5, 0.05, 0.01\}$. Every run memorised within 1 500 steps with test accuracy between 0.87 and 0.90, and the run at $\alpha = 0.001$ did not memorise.

Table 11: The MNIST runs setting by setting, beside the modular addition setting of Table 10. The learning-rate rule and the probe follow `mnist/README.md`.

Setting	MNIST	Modular addition
Task	1 000 training images drawn by data seed $42 + s$; all 10 000 test images	476 of 529 pairs, split by seed
Network	Three layers of width N, rectifier, no biases, standard parameterisation, uniform initial weights multiplied by s	Equation. (2), NTK parameterisation
Predictor	Uncentred, $f(x;\theta)$	Centred, $\alpha[f - f_0]$
Laziness	Initial scale $s \in \{3, 5, 7, 9\}$	$\alpha \in \{0.5, 1, 2\}$
Width N	$\{100, 200, 400, 800\}$	$\{50, \ldots, 1600\}$
Weight decay $\eta\lambda$	$\{0, 10^{-6}, 3\times10^{-6}, 6\times10^{-6}, 10^{-5}, 2\times10^{-5}, 4\times10^{-5}\}$, fixed per step	eight values from 0 to 3×10^{-3}
Optimiser and loss	SGD, batches of 200, no momentum; mean squared error on one-hot targets	Full-batch gradient descent
Learning rate	$\eta_0 = 1/\lambda_0$ with λ_0 the top Gauss-Newton eigenvalue at initialisation; reset every 10 000 steps to $\eta_0\lambda_0/\lambda_t$, capped at $1000\,\eta_0$	η_0/α^2, fixed
Budget and checkpoints	500 000 steps; 120 log-spaced steps and every 2 000 steps	200 000 steps; 307 checkpoints
Kernel and statistics	Closed-form kernel of the summed logits over $\sqrt{10}$, on a fixed draw of 1 000 test images	Kernel of the summed logits over $\sqrt{p}$, on the 53 test pairs
Events	Memorisation at 0.99; grokking at test accuracy 0.80, counted from step 0; censored at 500 000	Grokking at 0.95; censored at 200 000
Runs	560, all finished; 134 summed GPU hours on H100s	720

Table 12: Fits of (5) on MNIST at $N \in \{200, 400, 800\}$, with 95% intervals from the bootstrap of Table 3. The main row is the fit quoted in section 4.5: $s = 7$ with $\eta\lambda$ from 3×10^{-6} to 10^{-5}, where every seed groks. The other rows change one thing.

Fit	a	b	γ	σ
Main, $s = 7$, level 0.80 (45 runs)	-0.02 [-0.07, $+0.02$]	-1.00 [-1.05, -0.96]	$+0.01$ [-0.07, $+0.10$]	0.03
$s = 5$	-0.02 [-0.06, $+0.03$]	-1.00 [-1.04, -0.96]	$+0.01$ [-0.07, $+0.09$]	0.03
$s = 9$	-0.02 [-0.05, $+0.02$]	-1.00 [-1.02, -0.97]	$+0.00$ [-0.06, $+0.06$]	0.02
Adding $\eta\lambda$ up to 4×10^{-5}	-0.02 [-0.06, $+0.01$]	-1.00 [-1.01, -0.99]	$+0.03$ [-0.00, $+0.06$]	0.03
Level 0.85 (1 censored)	-0.08 [-0.15, -0.02]	-1.00 [-1.06, -0.94]	$+0.00$ [-0.11, $+0.15$]	0.03
Memorisation step	$+0.08$ [-0.02, $+0.15$]	-0.12 [-0.25, -0.04]	$+0.00$ [-0.21, $+0.23$]	0.09

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Indicate whether all your group members consent for your report to be used as a teaching resource by including one of the statement below.

We give consent for this to be used as a teaching resource.
We DO NOT give consent for this to be used as a teaching resource.

[TODO: Keep one of the two statements.]

Draft notes: delete before submission

1. How the report meets each marking criterion

The marking rubric (screenshot of 17 October 2026; the text that the screenshot cuts off was supplied by Jasper Chong) has five criteria. Each is worth 4 of the 20 marks of the report and is scored from 0 to 100%. The four bands are very good (75 to 100%), good (50 to 75%), poor (25 to 50%) and very poor (0 to 25%). The rubric regroups the five criteria of the task sheet (`project.pdf`). Presentation takes clarity and conciseness, correctness and reproducibility are now one criterion, and quality of work takes significance and novelty. Reference is new. The task sheet still sets the required content of the main text and its limit of 12 pages.

The first column quotes the rubric, first its short text and then its very good band. The other columns give where the report stands on 17 October 2026, the band we expect, and what would raise it. The bands are our own estimate.

Criterion (4 marks each)	Where the report stands	Likely band	What would raise it
Presentation. "Accurate language; organized, professional formatting according to the provided template; clear notation and technical statements." Very good: "Free of spelling, grammar, and punctuation errors. Content is well-organized and professionally formatted according to the provided template. Notation and technical statements are clear. Writing is concise and direct." (The screenshot cuts off the end of the short text.)	The preamble of the template is unchanged. One research question, and three statistics on one kernel (Equation. (3)). A cell is defined in section 3.2. Each experiment opens with its finding. The main text has five figures and five tables, with one colour for each statistic in every figure. The clear typos of the abstract and the Introduction were fixed on 17 October.	Good.	Write the Limitations as short prose; they are still a list under a TODO. Explain the accelerated failure time model for our audience (the TODO in section 3.5). Decide on the notation table. "Rises significantly" in the first paragraph of the Introduction may read as statistical significance. Check every figure at print size. Have a member who did not write a section read it for spelling and for the writing rules of CLAUDE.md. Remove all red markup.
Completeness. "Covers the problem, significance, solution, main findings, limitations of the study, and conclusion." Very good: "Clearly and adequately covers the problem, its significance, solution, main findings, limitations of the study, and conclusion."	The problem and its significance are in Sections 1 and 2, with Table 1. The solution is in Section 3, the findings in Section 4, the limitations in Section 5 and the conclusion in Section 6. The contribution, AI use and consent sections exist.	Good.	Section 4.5 (MNIST) and the Limitations were filled in on 20 October 2026. Fill in the contribution lines, the acknowledgement and the consent choice, and agree the AI-use text.

Criterion (4 marks each)	Where the report stands	Likely band	What would raise it
Correctness and reproducibility. "Sufficient technical details are provided. The work is reproducible and correct." Very good: "Work is technically correct and internally consistent. Methods, data, assumptions, and relevant implementation details are sufficiently specified to reproduce the results."	Table 10 gives every setting, the split rule and the compute. Appendix D gives every setting of the prediction models. `report_v2/code/` rebuilds every figure, table and number with `make_all.py` from a 9 MB extract with checksums, and every number in the text is a macro. The checks of 12 October recomputed the kernel by autograd, retrained the baseline run and reproduced 133 macros with independent code. The fits are censored, their intervals resample cells, and the prediction tests hold out whole cells or values of α.	Good to very good.	Add one check of the log-normal assumption, such as a probability plot of the residuals of the main fit. Say in the report where the code is, and submit it as a zip. Keep the gap in memorisation time against Kumar et al. stated (Appendix B); if that appendix goes, move one sentence to section 4.1.
Reference. "Citations support claims originating outside the project. References are professionally formatted, with no fake or hallucinated sources." Very good: "Claims originating outside the project are supported by appropriate citations. References are genuine, accurate, complete, professionally formatted, and clearly linked to the cited claims."	All 21 entries of `ref.bib` were checked against arXiv, Crossref or PubMed (Section 3 of these notes). Every section, equation and page number cited was checked in the version named in the text, and the text names that version.	Good.	Dyer and Gur-Ari (2020) is named in section 2 without a citation. Check it on arXiv and cite it, or drop the clause. Settle Khanh or Truong as the family name, and confirm the Kalbfleisch edition. The years follow the arXiv versions, as in the golden report: Jacot et al. 2020 for a NeurIPS 2018 paper and Cortes et al. 2024 for a JMLR 2012 paper. A marker may read these as wrong years. Adding the venue to these entries is a decision for the group.

Criterion (4 marks each)	Where the report stands	Likely band	What would raise it
Quality of work. "Significant and challenging project. Execution demonstrates technical clarity, rigor, creativity, and coherence, with well-motivated problems, well-justified decisions, and well-supported claims." Very good: "Project is significant and challenging. Execution demonstrates strong technical clarity, rigor, creativity, and coherence. Problems are well-motivated, decisions are well-justified, and claims are well-supported. Methods, findings, and conclusions form a coherent whole."	The question sets three recent accounts against each other, and Table 1 gives what each one predicts. Each tier answers one column of that table. The 720 runs cross laziness, width and weight decay. The new parts are the split of the kernel's movement into scale and rotation (Equation. (4)), the placebo test for threshold claims, the crossed grid, and prediction from measurable quantities against a settings reference. Every claim is held to the limits in `report_v2/CLAUDE.md`. The Conclusion states the answer.	Good.	Say in one sentence why a practitioner would measure S_t and R_t, since the settings reference still predicts better (Figure 5). State what a single small task can and cannot show. Search the literature beyond the works cited before any claim of a first. The MNIST tier of section 4.5 now answers the limit of one task. Decide whether the group accepts the curvature learning-rate rule as part of that setup (open decisions below).

Other requirements of the task sheet. The main text must end by page 12; on the build of 17 October 2026, with the red markup in, it ends on page 13, and the appendices run from page 14 to page 17. The group set the page limit aside for this round. The code goes to Blackboard as a separate zip, and each member submits a contribution statement.

2. Changelog: decisions and what was cut

Decisions taken on 5 October 2026.

- All results come from `data/`: 720 runs, 5 seeds, 200 000 steps, $\alpha \in \{0.5, 1, 2\}$, six widths and eight weight decays. The golden report used the grid-v0.1.0 databases (12 seeds, 100 000 steps) instead.
- The alignment A_t is the centred alignment of the summed-logit kernel on the test pairs (`A_sum` in `data/`). The golden report on main uses the tangent kernel of all p logits, after Baratin et al. The statistics S_t, R_t and A_t now share one kernel, and the prediction notebooks already use this A_t.
- The main text uses one grokking level, 0.95. The levels 0.80 and 0.90 move to Appendix E.
- The reproduction of Kumar et al. (2024) keeps a short section with one figure. The comparison setting by setting moves to Appendix B.
- Tier 4′ was first drafted as the rate-addition model. It was reopened the same day as a decision between five options (A to E), because that model rests on assumptions of our own.

- Each section gained a red draft contribution block that states, as if the preliminary findings hold, what is new and why it matters.
- The word "rotation" is kept for now and will become "shape" later.
- In Tier 3 the model given the settings is a reference, since a general network does not report its laziness or its effective regularisation.
- The planning checks changed one claim. The alignment does not pass the placebo test, so Tier 1 claims a threshold for rotation only.

Decisions taken on 11 October 2026 (Jasper Chong).

- Tier 4′ (Section 4.5, with its options A to E and its figure) is commented out in `main.tex`, and items E11, E17 and E18 were not run. To restore it, remove the leading comment sign from each line between the markers TIER 4' COMMENTED OUT and END OF TIER 4', and restore contribution 5 of the Introduction.
- (12 October 2026, Jasper Chong.) A new Section 4.5, Tier 4, holds a TODO placeholder for an extension experiment on MNIST digits. It is separate from the commented-out theory section Tier 4′. The MNIST runs would be new training outside `data/`.
- The spread in Tier 1 is the standard deviation, not the coefficient of variation. To restore the CV, set `SPREAD = 'cv'` in `report_v2/code/E03_placebo.py`; its JSON output keeps the CV values.
- Every figure, table and number now comes from the scripts in `report_v2/code/` through `numbers.tex`, `figures/` and `tables/`. The red planning values are gone except those of E14.
- The Tier 2 intervals resample whole cells and then seeds within cells. This settles the open decision on the intervals. Seed-only intervals are in `E06.json` for comparison.
- E14 (then Appendix D, dropped on 17 October 2026) was not rerun. It needs the grid-v0.1.0 databases and the R scripts of branch `feat/grid`.
- Kumar et al. App. 8.1 and 8.3, Sec. 3 and Fig. 2 (arXiv v3) were read for Appendix B. The memorisation steps differ from their figure; the grokking steps agree at $\alpha = 0.5$ and $\alpha = 1$.
- Claims changed by the full runs were marked in red with "Differs from draft" in the draft contribution blocks and the abstract. In short: the rotation threshold at a fixed width is not sufficient; the interval of γ includes zero; weight decay is one of the settings that set the scale; across widths no statistic passes only at level 0.95. These markers were removed on 12 October 2026 (below).

Changes made on 12 October 2026.

- The red draft contribution blocks were folded into the prose and deleted, together with the `draftcontrib` environment, the `\differs` macro and the commented block inside Tier 4′. Their text is in the git history of `main.tex` before this date. The abstract and the Conclusion were written from the final numbers.
- The "why it matters" and "what is new" points that the numbers support now sit in the Introduction (a paragraph after the contributions), Related Work (the Predictions paragraph and Kim's scope), section 4.1 (the gate and the baseline split), section 4.2 (the alignment hypothesis and the critical norm), section 4.3 (Kim's scope and the width-dependent decay edge) and section 4.4 (scoring, progress measure).

- Claims changed. (1) The Tier 2 claim and the Conclusion said that weight decay acts through the scale. Weight decay explains 0.26 of the spread of S_t, about as much as α and width, so the text now says that at a fixed width weight decay shrinks the scale. The story in `report_v2/CLAUDE.md` still says weight decay sets the scale. (2) The baseline claim "rotates by a few per cent" became "R_t stays below 0.09", the bound on every baseline checkpoint. (3) The reconciling claim of the Introduction block was dropped.
- Claims added from the final numbers. Tier 1 now reports that S_t pooled over all runs that grok tightens 1.69-fold, with the caveat that it keeps growing. Tier 3 now states in the prose that 0.96 to 0.99 of the spread lies between cells, so the statistics do not separate the seeds of a cell, and its claim is limited to held-out cells.
- The Introduction no longer carries a "search the literature" TODO. No sentence claims a first. The search is listed under novelty in the criteria table.
- The criteria table was rewritten against the five criteria and the required content of the task sheet.
- The red TODO around the interval sentence of section 3.5 was removed, since the intervals resample cells (decision of 11 October).
- Page limit. Built with the red markup hidden, the main text ended five lines into page 13. Words that repeat other sections were cut: Kim's scope in section 4.3 (kept in Related Work), a closing sentence of section 4.1, a scoring sentence of section 4.4 (kept in section 3.5), and parts of the Introduction and Conclusion. The surface figure (E9, commented out on 17 October) went from 0.66 to 0.6 of the line width. Two lines still spill onto page 13. The next cuts would be to write the Limitations as prose, or to merge Figure 4 into Figure 3.
- The code and the numbers were verified. The tangent kernel recomputed by autograd from the saved weights of 8 runs matches S_t, R_t and A_t to a relative 3×10^{-10}. A minimal retraining of the baseline run reproduces its accuracy curve exactly over 6 000 steps. Independent code reproduces 133 macros of `numbers.tex`; the only gaps are Monte Carlo ones (a permutation p and a bootstrap bound, both within their noise). A clean rerun of `make_all.py` reproduces `numbers.tex`, every table and every JSON output exactly.
- One claim was corrected. The text said that at levels 0.80 and 0.90 the scale and the alignment tighten across widths. At 0.90 the scale has $p = 0.085$, so only the alignment does. Section 4.2 and section E now say so.

Changes made on 17 October 2026 (Jasper Chong's notes).

- Tier 3′ is renamed Tier 3 everywhere in `report_v2/`: in this file, `report_v2/CLAUDE.md`, `code/README.md` and `code/models.py`. The commented-out theory section Tier 4′ keeps its name.
- Each experiment section opens with "Finding." in place of "Claim.".
- Section 4.1 was cut to the result, the figure and the split of D_t^2 at grokking for $\alpha = 1$. The paragraph of event-step medians was dropped, since Table 6 gives them. The gap in memorisation time and the rest of the split moved to Appendix B. A TODO there asks whether the report needs that appendix. The earlier text is in the git history of `main.tex` before this date.
- A cell is now defined in section 3.2, and the captions of Table 10 and Figure 3 use the same wording.

- Figure E5 (the grokking time against weight decay at each width) is commented out of section 4.3, with a TODO to consider it for the seminar. To restore it, remove the comment signs below the marker FIGURE E5 COMMENTED OUT.
- Tier 3 was simplified. It now opens with the column of Table 1 that it answers and runs two tests, held-out cells (Table 4) and an unseen α (Figure 5). Figure E8 was redrawn by `code/E08_ladder.py` with five bars: each statistic alone, the trees on all three at t^* and the recurrent network on their history. Its overlapping label and its cut-off axis label were fixed. The linear model on all three, the model with interactions and kernel ridge are no longer in the figure or the text, but `E08.json` keeps their scores. The rerun reproduced `E08.json` exactly, and `numbers.tex` did not change. Section 3.5 and Appendix D were trimmed to match.
- Figure E9 (the surface over S_t and R_t) and the product coefficient γ_{SR} are commented out of section 4.4. The claim that the best models let rotation and scale interact was dropped from the finding. The comment block below the marker FIGURE E9 AND THE PRODUCT COEFFICIENT COMMENTED OUT holds the old text and says how to restore it.
- Typos fixed without changing the meaning: "phenomena" to "phenomenon", the stray "by" and the missing full stop of the abstract, the missing article of the Introduction's second paragraph, contributions 2 and 3, "sole" to "single", the last sentence of the Introduction, the gloss of "weight-decay clock" in section 2, and the sentence on SDs in section 3.5.
- The criteria table was rewritten against the marking rubric.
- At Jasper Chong's request, Tier 3 does not test the weight decay as a predictor, and the report now says why. We set $\eta\lambda$, while the effective regularisation of a general network cannot be measured, and the report looks for predictors that carry over to real models. The caption of Table 1, the Prediction paragraph of section 3.5 and the opening of section 4.4 each say so. `code/E08_ladder.py` now also fits the trees to $\log N$ alone and to $\eta\lambda$ alone. Both rank runs worse than the two settings together, and section 4.4, its finding and the Conclusion use this as support for the mixed clock. The rerun reproduced every earlier value of `E08.json`. An earlier draft of the same day put the weight-decay fits (kernel ridge on $\eta\lambda$ alone and with the statistics, and the trees on $\eta\lambda$ with $\log R_t$ or S_t) in a new Appendix D. That appendix and its code were removed as confusing. They are not in the git history, so they would need to be rewritten.
- Appendix D, the cross-check from the second pipeline (E14), was dropped. It held only a TODO that quoted the golden report. Its text is in the git history of `main.tex` before this date.

Changes made on 20 October 2026 (Samuel Rieger).

- Tier 4 (section 4.5) now holds the MNIST extension, built from the sweep of 19 October 2026 in `mnist/sweep/` (560 runs, trained by Zac Kienzle on the H100s). Its figure, table and numbers come from `report_v2/code/E19_mnist.py`, which reads `mnist/sweep/` from the repository. This is the one exception to the rule that every result comes from `data/`; `report_v2/CLAUDE.md` records it.
- New: section 3.6 (the second task, in Methods), Appendix G (its settings and the check rows of its fit), a contribution in the Introduction, a sentence on Liu et al. (2023) in Related Work, and that paper in the weight-decay row of Table 1.

- Claims changed. The abstract, the end of the Introduction and the Conclusion no longer say that rotation matters most or that each clock describes one part of the time. They say that weight decay sets the time through the weight norm on both tasks, and that width enters only on modular addition, where the kernel must change shape before the task can be solved. The second half of that is our reading, and the text says so.
- To make room: the Limitations are prose; the paragraph on seed-only intervals moved from section 4.3 to Appendix E; the opening paragraph of section 4.4 lost two sentences. Every change is in blue with a bracketed note.
- The MNIST events keep the report's definitions, except that the grokking level is 0.80 and the budget is 500 000 steps, since no MNIST run reaches 0.95. section 3.6 says so.
- Later on 20 October: the MNIST text was reworded to say how the delay arises there, following Liu et al. (2023, Secs. 2, 4 and 5): the enlarged initial function is kept by the uncentred predictor and weight decay must shrink it, whereas the centred predictor of section 3.1 removes it. section 3.6 cites the group's pilot of 12 October 2026 (`mnist/exploration/results/mnist_pilot/`, one seed), in which the centred predictor on MNIST showed no delay at any α. Decide whether a one-seed pilot may be cited in the main text.
- Also later on 20 October, for the tutor's feedback on the proposal (more datasets, stated assumptions, expected outcomes): the Introduction says why a second task was added, the Predictions paragraph of section 2 states what each clock predicts for MNIST, the Tier 4 finding recalls those predictions, and the Limitations state the assumption behind the comparison.
- Last on 20 October, for the page limit: the MNIST text was cut to the two points that matter, how the delay is induced on MNIST against modular addition, and what that does to the Tier 1 and Tier 2 tests. The abstract, contribution, Related Work, section 3.6, section 4.5, Limitations and Conclusion are shorter; Figure 6 is narrower. Nothing was dropped from the record: the check rows, width 100, the no-decay runs, the alignment, the placebo ratios and the pilot are in Appendix G. The entry for Liu et al. (2023) in `ref.bib` has a `shortauthor` field so that it cites as "Liu et al."; the other three-author entries still cite in full, which `maxcitenames=2` in the preamble would change for all of them.
- Panel (c) of Figure 6 is now the time and outcome map of Figure 3 at $s = 7$, in place of rotation against width. Drawing it showed that width 100 does reach the level in most runs with decay, so the caption, the Limitations and Appendix G no longer say that it never groks; they say it is later than the wider networks, with little delay to time, and off the law.
- A second cut of about 16 lines from the MNIST additions: the MNIST predictions sentence in section 2 became the clause "on either task" before Table 1; the pure-decay check, the placebo reason, the $s = 3$ accuracy, the check-row sentence and the note on Tier 3 moved from section 4.5 to Appendix G; section 3.6, the Limitations paragraph and the Conclusion lost a clause each. The two-route contrast, the finding and the width-100 caveat are unchanged.
- The Limitations and the Conclusion were re-marked so that only the MNIST additions are blue. Limitations paragraph 1 is the group's bullet list as prose, word for word, with "modular addition" added; the proposal items are back in that paragraph. The Conclusion keeps the original sentences, with "On modular addition" before "rotation matters most", the MNIST sentences in place of "Each published clock therefore describes one part of

it", "on two tasks" after "the crossed grid", and the frozen-kernel test added to the next step.

- The abstract was rewritten at about half its length, with the group's permission: one sentence per result, no numbers, the MNIST result in two sentences, and the commented-out line removed.
- To save space, Table 10 moved from section 3.1 to a new Appendix F just before the MNIST appendix, and Figure 3 and Figure 5 were narrowed from 0.72 to 0.56 of the line width. Nothing in the table or the figures changed.

Cut from the golden report. Each item can be restored from the golden report's source, `docs/report_golden.tex` (added in commit 7da1996), and its figure files in `docs/figures/`.

- Section 3.4, Spectral measures: effective rank, trace ratios and the target power along eigenvectors. Section 4.3: the cumulative target power figure (`docs/figures/spectrum.tex`).
- Section 3.5, Event times: the Kaplan-Meier estimate with Greenwood's interval, the phase labels, the additive model of Aalen and the relative risk model with time-dependent covariates. Section 4.3: the table of that model. Appendix B, Model checks: the probability plot and the cumulative regression functions (`docs/figures/probability.tex`, `aalen.tex`).
- Section 3.6, Choice of levels: optimal designs, the censored information matrix, the broken line and the sequential design. Section 4.2.2: the scan, the design table, the broken-line figure, the survival figure and the accelerated failure time table of the weight-decay arm (`docs/figures/phases.tex`, `broken.tex`, `survival.tex`). A short summary stood in Appendix D until it was dropped on 17 October 2026.
- Section 4.2.1: the α sweep with $\alpha = 1.5$ and 12 seeds (`docs/figures/alpha_sweep.pdf`).
- Section 3.2: the first-logit kernel as a second output.
- Related Work: the paragraph on kernel alignment and spectral bias, silent alignment, the early and late dichotomy, Omnigrok, and provable grokking in ridge regression.
- The pipeline itself: branch `feat/grid`, folder `grid/` (Snakemake workflow, DuckDB databases, R scripts).

Planned in the proposal and not run. The clamped-norm arm with $\rho \in [1, 2]$, the arm over three primes and three training fractions, the cross-entropy arm, the statistic $y^\top K^{-1} y$, and events defined by loss thresholds. The proposal is `docs/G2_proposal.tex`.

Superseded analyses. `repoduced-code/tier1_v2.ipynb` and `tier2_width_decay.ipynb` use the old data, with one split shared by every seed, the trace kernel and test accuracy 1.0. Their methods are ported, and their numbers are not used.

Open decisions for the group.

- Keep the title, or change it to give the answer.
- Keep Figure 4 as its own figure, or merge it into Figure 3.
- Whether to add the notation table.
- Which theory direction, if any, Tier $4'$ takes (options A to E in the commented-out Section 4.5 of `main.tex`).

- Confirm the softer Tier 2 headline of 12 October 2026 ("at a fixed width, weight decay shrinks the scale"), and update the story in `report_v2/CLAUDE.md` to match.
- Whether the report needs Appendix B for the reproduction (17 October 2026).
- Whether to use figure E5 (the grokking time against weight decay at each width) in the seminar (17 October 2026).
- Whether the group accepts the curvature learning-rate rule of the MNIST runs (section 3.6) as part of the setup. Every strong MNIST cell uses it. The only fixed-rate grokking in the exploration was weak, at $s = 3$ (20 October 2026).
- Whether to keep the reading that width enters only where the task forces the kernel to change shape (section 4.5 and the Conclusion). It rests on Mohamadi et al. (2024) and is ours (20 October 2026).

3. Reference checks

Every entry of `ref.bib` was checked on 5 October 2026 against its arXiv abstract page, its DOI record on Crossref, or PubMed. Each section and equation number cited in the text was checked in the PDF of the version named.

Entry	Checked against	Status	Notes
Power et al. 2022	arXiv 2201.02177	Matches	v1 only.
Nanda et al. 2023	arXiv 2301.05217	Matches	v3 of 19 Oct 2023 gives the month. The v3 PDF was read on 12 Oct 2026 for the progress-measure sentence of section 4.4. Page 1 defines progress measures as "metrics that precede and are causally linked to the phase transition, and which vary more smoothly", after Barak et al. (2022). The report claims only the first and last conditions for R_t.
Gromov 2023	arXiv 2301.02679	Matches	
Jacot et al. 2020	arXiv 1806.07572	Matches	Year is v4 (Feb 2020); NeurIPS 2018.
Chizat et al. 2020	arXiv 1812.07956	Matches	Year is v5 (Jan 2020); NeurIPS 2019.
Kumar et al. 2024	arXiv 2310.06110	Matches	v3 of 11 Apr 2024. App. 8.1 and 8.3, Sec. 3 and Fig. 2 read in the v3 PDF on 11 Oct 2026 (Appendix B).
Mohamadi et al. 2024	arXiv 2407.12332	Matches	ICML 2024.
Mohamadi, Bae, Sutherland 2022	arXiv 2206.12543	Matches, entry type changed	The golden bib types it as an article with no journal. It is typed misc here. Published at ICML 2023.

Entry	Checked against	Status	Notes
Lewkowycz, Gur-Ari 2021	arXiv 2006.08643, PDFs of v1 and v2	Matches	Year is v2 (Jan 2021); NeurIPS 2020. CLAUDE.md gives both 2020 and 2021. The kernel flow is Eq. S5 and T_t is Eq. S6 in both v1 and v2. The proposal's draft note, which says they become S8 and S9 in v2, is wrong: S8 in v2 is the infinite-width solution. $E[T_0] = O(n^{-1})$ is credited to Dyer and Gur-Ari (2020), which is not in ref.bib.
Fort et al. 2020	arXiv 2010.15110	Matches	NeurIPS 2020.
Geiger et al. 2020	Crossref and arXiv 1906.08034	Matches	JSTAT 2020, 113301.
Cortes et al. 2024	arXiv 1203.0550	Matches	Year is v3 (Apr 2024). The paper is JMLR 13 (2012), 795–828.
Khanh et al. 2026	arXiv 2606.13753	Matches arXiv; name order flagged	arXiv lists "Truong Xuan Khanh" and parses the surname as Khanh, so "Khanh et al." matches arXiv. In Vietnamese order the family name is usually written first, which would make it "Truong et al.". The co-authors have the same question. The proposal's key is `truong2026norm`. The group should decide.
Khanh 2026	arXiv 2606.18465	Matches arXiv; name order flagged	As above.
Kim 2026	arXiv 2607.23967	Matches	Cor. 5 and Eq. 20 checked in the v1 PDF (page 8). The Scope paragraph cited in Section 2 and Tier 2 is on page 3 of v1. It lists grokking without explicit weight decay and the creation of new population-visible directions by feature learning among what the mechanism does not explain.
Pracher et al. 2026	arXiv 2609.26679	Matches, fields added	The golden bib gives only the DOI. The eprint, month and class were added. v1 of 22 Sep 2026.
Kalbfleisch, Prentice 2002	Crossref, Open Library	Matches, edition changed	The golden bib gives edition 1. The 2002 Wiley printing is the second edition; the first appeared in 1980. Crossref lists the online version as edition 1, so this needs a check of the book itself.
Davidson-Pilon 2019	Crossref	Matches	JOSS 4(40), 1317.
Harrell et al. 1982 (new)	PubMed 7069920, Crossref	Matches	JAMA 247(18), 2543–2546. Crossref lists only the first author; PubMed lists all five.
Chen, Guestrin 2016 (new)	arXiv 1603.02754	Matches	KDD 2016.

Entry	Checked against	Status	Notes
Cho et al. 2014 (new)	arXiv 1406.1078	Matches	EMNLP 2014.
Liu, Michaud, Tegmark 2023 (new, 20 Oct 2026)	arXiv 2210.01117, PDF of v2	Matches	v2 of 23 Mar 2023; ICLR 2023. The entry is the golden report's, without its abstract field. Section 2 (page 3) gives $w(t) \approx e^{-\gamma t} w_0$ and $t \approx \ln(w_0/w_c)/\gamma$ when weight decay is the only regulariser and the training loss is negligible after fitting; Figure 2(c) shows the time to 95% test accuracy inversely proportional to the decay. Section 4 (page 5) gives the MNIST recipe: 1k training images, Kaiming uniform initial weights scaled by $\alpha > 1$, a depth-3 width-200 ReLU MLP, AdamW, MSE on one-hot targets. Our runs use SGD without biases in place of AdamW.